

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular End Semester Examination – Summer 2022**

**Course: B. Tech.**

**Branch : AI&DS**

**Semester : IV**

**Subject Code & Name: BTAIC401 Data Analysis**

**Max Marks: 60**

**Date: 12/08/2022**

**Duration: 3.45 Hr.**

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.
5. Use of Z-Table, T-Table and F-Distribution table is allowed.

|                                                                                                                                                                                                                                                                                                                                                                                                                                                    | (Level/CO) | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                        |            |       |
| A) Describe PPDAC: The Statistical Method in detail.                                                                                                                                                                                                                                                                                                                                                                                               | (BT2/CO1)  | 6     |
| B) Define Missing data and Explain techniques and tools to handle Missing data.                                                                                                                                                                                                                                                                                                                                                                    | (BT2/CO1)  | 6     |
| C) Discuss the following <ol style="list-style-type: none"><li>i. Different Techniques in Statistical Modelling</li><li>ii. Different types of BIAS</li></ol>                                                                                                                                                                                                                                                                                      | (BT2/CO1)  | 6     |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                         |            |       |
| A) Discuss Measure of Central Tendency- Mean, Mode, and Median with suitable example.                                                                                                                                                                                                                                                                                                                                                              | (BT2/CO2)  | 6     |
| B) <ol style="list-style-type: none"><li>i. List and explain Different Measure of Spread or Dispersion with suitable example.</li><li>ii. Identify/Find the Variance and Standard Deviation of the Following Numbers: 1, 3, 5, 5, 6, 7, 9, 10.</li></ol>                                                                                                                                                                                           | (BT1/CO2)  | 6     |
| C) Compare different types of Skewness.                                                                                                                                                                                                                                                                                                                                                                                                            | (BT4/CO2)  | 6     |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                        |            |       |
| A) Define Transformations and explain how Box-Cox Power Transformation works.                                                                                                                                                                                                                                                                                                                                                                      | (BT2/CO3)  | 6     |
| B) Describe Square Root Transform with suitable example.                                                                                                                                                                                                                                                                                                                                                                                           | (BT2/CO3)  | 6     |
| C) Explain What Normal Transform is? State how it is used for standardization?                                                                                                                                                                                                                                                                                                                                                                     | (BT4/CO3)  | 6     |
| <b>Q.4 Solve Any Two of the following.</b>                                                                                                                                                                                                                                                                                                                                                                                                         |            |       |
| A) Apply independent samples T-Test for the following data sets:<br>Data set A: 1,2,2,3,3,4,4,5,5,6<br>Data set B: 1,2,4,5,5,5,6,6,7,9                                                                                                                                                                                                                                                                                                             | (BT3/CO4)  | 6     |
| B) <ol style="list-style-type: none"><li>i. Enlist and Explain different types of Z-Test.</li><li>ii. A survey claims that 9 out of 10 doctors recommend aspirin for their patients with headaches. To test this claim, a random sample of 100 doctors is obtained. Of these 100 doctors, 82 indicate that they recommend aspirin. Is this claim accurate? Use <math>\alpha = 0.05</math>.<br/>(Calculate Result using Appropriate Test)</li></ol> | (BT3/CO4)  | 6     |
| C) Describe G contingency table test with suitable example.                                                                                                                                                                                                                                                                                                                                                                                        | (BT2/CO4)  | 6     |

**Q. 5 Solve Any Two of the following.**

**A) Interpret result using one way ANOVA test on following scenario. (BT3/CO5) 6**

A researchers wants to test a new anxiety medication. They split participants into 3 conditions (0MG, 50MG and 100 MG), Then ask them to rate their anxiety level on a scale of 1-10. Are there any differences between the three conditions using  $\alpha=0.05$ ?

| 0MG | 50MG | 100 MG |
|-----|------|--------|
| 9   | 7    | 4      |
| 8   | 6    | 3      |
| 7   | 6    | 2      |
| 8   | 7    | 3      |
| 8   | 8    | 4      |
| 9   | 7    | 3      |
| 8   | 6    | 2      |

**B) Describe Mood's median Test with suitable example. (BT2/CO5) 6**

**C) Discuss the following (BT2/CO5) 6**

- MANOVA Test
- ANCOVA Test

**\*\*\* End \*\*\***